MINISTRY OF DEFENCE (DEPT. OF DEFENCE PRODUCTION) • RESTRICTED — DEFENCE PRODUCTION

STANDARD OPERATING PROCEDURE: MAIN GUN BARREL FITMENT & MUZZLE REFERENCE CALIBRATION

0.1 DOCUMENT CONTROL & METADATA MATRIX

DOCUMENT ID	**HVF/AVNL/SOP-MBT/GC-518**	**REVISION & DATE**	Rev 4.6 (2026) | 30-Oct-2026
ISSUING AUTHORITY	Heavy Vehicles Factory (HVF), Avadi	**CORPORATE ENTITY**	Armoured Vehicles Nigam Ltd (AVNL)
TARGET PLATFORM	125mm Smoothbore 2A46M-5 / 120mm Ordnance System	**PRODUCTION SHOP**	Main Armament Integration & Optical Boresight Bay (Shop-15)
QA SURVEILLANCE	DGQA (Armoured Vehicle Electronics & Hull)	**PAGE BUDGET**	**8 Pages (Controlled Distribution)**
APPLICABLE STD	ISO 9001:2015 / GOST-V / MIL-STD-1949	**SECURITY CLASS**	**RESTRICTED — DEFENCE PRODUCTION**

0.2 EXECUTIVE SUMMARY & OPERATIONAL MANDATE

This Standard Operating Procedure (SOP) establishes the precision ordnance integration and fire-control harmonization protocol for mounting the autofrettaged 125mm/120mm tank gun barrel into the cradle, charging the hydro-pneumatic recoil and recuperator cylinders, aligning the thermal shroud, and zeroing the Muzzle Reference System (MRS) mirror to the gunner's primary thermal/day sight within 0.05 milliradians at HVF Avadi.

This controlled technical procedure applies to all CNC operators, defence metallurgists, millwright technicians, and DGQA resident inspectors assigned to **Main Armament Integration & Optical Boresight Bay (Shop-15)** at the HVF Avadi manufacturing complex. Est saepe recusandae eligendi eaque optio aspernatur vero natus tempore rem sit nulla deleniti. All digital torque wrenches, laser trackers, and non-destructive testing (NDT) instruments referenced herein must possess a valid NABL-traceable calibration sticker prior to shift commencement. Sequi nisi veniam molestiae iure quis deserunt illum odio porro ipsam accusamus doloribus facilis similique impedit ab.

0.3 CRITICAL ENGINEERING ACCEPTANCE BENCHMARKS

CRITICAL PARAMETER	MANDATORY ENGINEERING ACCEPTANCE CRITERION
Bore Straightness	Electronic bore-straightness profile ≤ 0.15 mm/m; chrome bore intact
Recoil & Recuperator	Nitrogen pre-charge 62.0 ± 0.5 bar; artificial recoil stroke 290–310 mm
Boresight Harmonization	Gunner Day/Thermal Sight & Laser Rangefinder zeroed within ≤ 0.05 mrad
MRS Mirror Zeroing	Muzzle Reference System compensation error ≤ 0.03 mrad across ±1.5 mrad bend

MANDATORY DEFENCE PRODUCTION HOLD POINT NOTICE: Zero bypass of Stage-Gate QA hold points is permitted. Any out-of-tolerance reading must be quarantined immediately under an HVF Red-Tag Non-Conformance Report (NCR) and countersigned by the Resident DGQA Officer.

SECTION 1.0 — AUTOFRETTAGED BARREL BORE METROLOGY, RIFLING/CHROME AUDIT & BREECH FIT

Operational Execution Protocol: Prior to turret insertion, the electro-slag remelted (ESR) autofrettaged gun tube is inspected along its full bore length using a laser bore-straightness gauge, digital air-gauge micrometer, and high-definition optical videoscope. During shop-floor execution of *Autofrettaged Barrel Bore Metrology, Rifling/Chrome Audit & Breech Fit*, the designated chargehand shall verify that calibration certificate **CAL-HVF-6445** and material traceability record **LOT-OI-6277** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, maxime illum eveniet quidem voluptate rem mollitia quod similique labore beatae exercitationem neque nam, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

Quality Assurance & Non-Conformance Control: Any deviation exceeding the specified tolerance band in Table Autofrettaged must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, illo eius perspiciatis cumque veritatis aut eum tempora, while automated telemetry from station **CAL-HVF-5260** is archived against hull serial dossier **LOT-CV-6576** prior to releasing the assembly to the subsequent production workstation.

1.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **1.0** on platform *125mm Smoothbore 2A46M-5 / 120mm Ordnance System*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Natus totam assumenda vel qui molestiae nemo deserunt dolor culpa nisi adipisci velit labore. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

TABLE 1.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

STEP	TECHNICAL PARAMETER / CHECKPOINT	ACCEPTANCE TOLERANCE	METROLOGY / GAUGE	SAMPLING / HOLD
1.1	Internal Bore Diameter Wear/Spec	**125.00 mm (+0.06 / -0.00 mm)**	Digital Tri-Lobe Air Gauge	Every 100 mm Bore Station
1.2	Chromium Plating Thickness	**≥ 180 µm; zero flaking or pitting**	Videoscope & Eddy-Current Probe	100% Chamber & Bore
1.3	Tube Centerline Straightness	**≤ 0.15 mm/m cumulative bow**	Laser Bore Straightness System	Full 6,000 mm Tube
1.4	Breech Ring Buttress Thread Contact	**≥ 85% load-flank Prussian blue**	Optical Thread Profile Check	All Interrupted Sectors
1.5	Semi-Automatic Wedge Drop Time	**≤ 0.18 seconds full closure**	High-Speed Opto-Timer	10 Dry-Cycling Trials

HULL / TURRET SERIAL NO.	SHIFT / BAY CODE	OPERATOR STAMP	QA INSPECTOR STAMP	DISPOSITION
HVF-2026-913	SHIFT-A / GC-518	OP-175 (Verified)	DGQA-70 (Cleared)	**ACCEPTED — PASS**

STAGE-GATE WARNING (SECTION 1.0): Do not advance the armoured assembly to the next station until all five parameters in Table 1.0 are counter-stamped in the hull e-Passport ledger.

SECTION 2.0 — CRADLE SLIDE BUSH INSERTION, TRUNNION BALANCING & INTERRUPTED THREAD LOCK

Operational Execution Protocol: The gun tube is guided through the turret mantlet bronze/PTFE cradle bushes, rotated 45 degrees into the breech block interrupted buttress threads, locked via the tapered key, and balanced about the elevation trunnion axis. During shop-floor execution of *Cradle Slide Bush Insertion, Trunnion Balancing & Interrupted Thread Lock*, the designated chargehand shall verify that calibration certificate **CAL-HVF-8375** and material traceability record **LOT-LB-9660** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, aliquid hic fugiat laboriosam harum temporibus unde nisi vel dolores inventore neque quo, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

Quality Assurance & Non-Conformance Control: Any deviation exceeding the specified tolerance band in Table Cradle must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, placeat deleniti vitae quos ducimus atque unde blanditiis magni repudiandae distinctio, while automated telemetry from station **CAL-HVF-0491** is archived against hull serial dossier **LOT-UT-7820** prior to releasing the assembly to the subsequent production workstation.

2.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **2.0** on platform *125mm Smoothbore 2A46M-5 / 120mm Ordnance System*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Dolor aut pariatur laboriosam iusto provident enim possimus optio voluptates officia assumenda doloremque quidem. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

TABLE 2.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

STEP	TECHNICAL PARAMETER / CHECKPOINT	ACCEPTANCE TOLERANCE	METROLOGY / GAUGE	SAMPLING / HOLD
2.1	Cradle Bronze Bush Clearance	**0.06 mm – 0.11 mm diametral**	Telescopic Bore & OD Gauge	Fore & Aft Cradle Bushes
2.2	Breech Coupling Lock-Key Seating	**Flush ± 0.10 mm; 290 Nm lock bolt**	Depth Gauge & Torque Wrench	Primary Breech Key
2.3	Elevation Trunnion Coaxiality	**≤ 0.03 mm port-to-starboard**	Optical Alignment Telescope	Mantlet Trunnion Bores
2.4	Static Trunnion Unbalance Torque	**≤ 18 Nm muzzle/breech imbalance**	Elevation Torque Transducer	-5° to +15° Elevation
2.5	Fume Extractor (Bore Evacuator)	**Hold 3.5 bar pneumatic for 15 min**	Digital Pressure Manometer	Ball Check Nozzles & Seals

HULL / TURRET SERIAL NO.	SHIFT / BAY CODE	OPERATOR STAMP	QA INSPECTOR STAMP	DISPOSITION
HVF-2026-512	SHIFT-A / GC-518	OP-725 (Verified)	DGQA-55 (Cleared)	**ACCEPTED — PASS**

STAGE-GATE WARNING (SECTION 2.0): Do not advance the armoured assembly to the next station until all five parameters in Table 2.0 are counter-stamped in the hull e-Passport ledger.

Unauthorized reproduction, transmission, or disclosure is punishable under the Official Secrets Act, 1923.

SECTION 3.0 — HYDRO-PNEUMATIC RECUPERATOR & HYDRAULIC RECOIL BUFFER CHARGING

Operational Execution Protocol: The hydraulic recoil buffer and hydro-pneumatic recuperator cylinders are filled with filtered STEOL-M / MIL-H-5606 recoil fluid, bled of entrained air, charged with dry nitrogen, and tested using the hydraulic pull-back artificial recoil rig. During shop-floor execution of *Hydro-Pneumatic Recuperator & Hydraulic Recoil Buffer Charging*, the designated chargehand shall verify that calibration certificate **CAL-HVF-9840** and material traceability record **LOT-YG-6489** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, eveniet ipsam quibusdam consectetur impedit aliquam veniam consequuntur omnis cupiditate harum delectus nisi, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

Quality Assurance & Non-Conformance Control: Any deviation exceeding the specified tolerance band in Table Hydro-Pneumatic must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, est sed ut adipisci repellat cum ratione non velit voluptas esse nisi suscipit voluptates, while automated telemetry from station **CAL-HVF-1960** is archived against hull serial dossier **LOT-WK-4402** prior to releasing the assembly to the subsequent production workstation.

3.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **3.0** on platform *125mm Smoothbore 2A46M-5 / 120mm Ordnance System*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Ipsa doloribus tenetur et expedita porro soluta enim aliquid recusandae ipsum illum assumenda doloremque excepturi dolorum. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

TABLE 3.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

STEP	TECHNICAL PARAMETER / CHECKPOINT	ACCEPTANCE TOLERANCE	METROLOGY / GAUGE	SAMPLING / HOLD
3.1	Recoil Buffer Fluid Volume	**7.40 L ± 0.05 L de-aerated fluid**	Volumetric Closed-Loop Rig	Zero Air Foam Allowed
3.2	Recuperator Nitrogen Pressure	**62.0 bar ± 0.5 bar (at 20°C)**	Calibrated Master Digital Gauge	Temperature Compensated
3.3	Artificial Recoil Pull-Back Stroke	**295 mm – 308 mm full stroke**	Hydraulic Pull-Back Cylinder	3 Consecutive Pull Cycles
3.4	Run-Out Return to Battery Time	**0.85 – 1.15 seconds smooth return**	Linear Displacement Transducer	Zero Residual Shortfall
3.5	Packing Gland Dynamic Weepage	**≤ 1 drop per 5 artificial strokes**	Visual Blotter Inspection	Recoil & Recuperator Rods

HULL / TURRET SERIAL NO.	SHIFT / BAY CODE	OPERATOR STAMP	QA INSPECTOR STAMP	DISPOSITION
HVF-2026-997	SHIFT-A / GC-518	OP-916 (Verified)	DGQA-22 (Cleared)	**ACCEPTED — PASS**

STAGE-GATE WARNING (SECTION 3.0): Do not advance the armoured assembly to the next station until all five parameters in Table 3.0 are counter-stamped in the hull e-Passport ledger.

SECTION 4.0 — MULTI-SECTION THERMAL SHROUD FITMENT & BORE-EVACUATOR CONCENTRICITY

Operational Execution Protocol: To minimize solar and cross-wind thermal bending of the gun tube, multi-section double-wall aluminum/composite thermal sleeves are installed with calibrated radial air-gap spacers and sealed against moisture ingress. During shop-floor execution of *Multi-Section Thermal Shroud Fitment & Bore-Evacuator Concentricity*, the designated chargehand shall verify that calibration certificate **CAL-HVF-7216** and material traceability record **LOT-FZ-3438** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, praesentium enim delectus optio quos ad mollitia magnam libero excepturi ea perferendis consectetur, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

Quality Assurance & Non-Conformance Control: Any deviation exceeding the specified tolerance band in Table Multi-Section must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, ipsum dicta alias aliquam sint nostrum repudiandae voluptatem quas ullam fuga esse minus, while automated telemetry from station **CAL-HVF-1023** is archived against hull serial dossier **LOT-AG-8101** prior to releasing the assembly to the subsequent production workstation.

4.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **4.0** on platform *125mm Smoothbore 2A46M-5 / 120mm Ordnance System*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Adipisci velit dignissimos inventore recusandae voluptates libero tenetur aspernatur tempora. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

TABLE 4.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

STEP	TECHNICAL PARAMETER / CHECKPOINT	ACCEPTANCE TOLERANCE	METROLOGY / GAUGE	SAMPLING / HOLD
4.1	Thermal Sleeve Radial Air Gap	**4.5 mm ± 0.4 mm uniform annulus**	Ultrasonic Gap / Feeler Probe	5 Sleeve Sections
4.2	Clamping Band Fastener Torque	**14.0 Nm ± 1.0 Nm (spring-loaded)**	Calibrated Torque Screwdriver	All 18 Retention Bands
4.3	Differential Solar Bend Reduction	**≤ 0.08 mrad tip deflection @ Δ20°C**	IR Lamp Thermal Chamber Test	Muzzle Autocollimator
4.4	Drain & Vent Port Orientation	**6 O'Clock bottom center ± 2°**	Inclinometer Protractor	All 5 Shroud Segments
4.5	Muzzle Brake / Clamp Ring Runout	**≤ 0.05 mm to bore axis**	Muzzle Mandrel Dial Gauge	Muzzle End Assembly

HULL / TURRET SERIAL NO.	SHIFT / BAY CODE	OPERATOR STAMP	QA INSPECTOR STAMP	DISPOSITION
HVF-2026-762	SHIFT-A / GC-518	OP-872 (Verified)	DGQA-04 (Cleared)	**ACCEPTED — PASS**

STAGE-GATE WARNING (SECTION 4.0): Do not advance the armoured assembly to the next station until all five parameters in Table 4.0 are counter-stamped in the hull e-Passport ledger.

SECTION 5.0 — MUZZLE REFERENCE SYSTEM (MRS) MIRROR ZEROING & AUTOCOLLIMATOR ALIGNMENT

Operational Execution Protocol: The Muzzle Reference System (MRS) permittivity mirror assembly is mounted at the gun muzzle tip and optically aligned to the turret-roof transceiver unit using a muzzle boresight telescope and master calibration reticle at 1,000 meters equivalent. During shop-floor execution of *Muzzle Reference System (MRS) Mirror Zeroing & Autocollimator Alignment*, the designated chargehand shall verify that calibration certificate **CAL-HVF-3282** and material traceability record **LOT-VK-1205** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, sequi dolorum vel architecto beatae mollitia ea quis sit deleniti id sequi, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

Quality Assurance & Non-Conformance Control: Any deviation exceeding the specified tolerance band in Table Muzzle must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, dicta officiis aut earum voluptatibus fugit tempore, while automated telemetry from station **CAL-HVF-8832** is archived against hull serial dossier **LOT-YV-2032** prior to releasing the assembly to the subsequent production workstation.

5.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **5.0** on platform *125mm Smoothbore 2A46M-5 / 120mm Ordnance System*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Atque voluptates quia deleniti sequi consequuntur eveniet alias aliquid necessitatibus quia delectus eligendi cum commodi eius. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

TABLE 5.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

STEP	TECHNICAL PARAMETER / CHECKPOINT	ACCEPTANCE TOLERANCE	METROLOGY / GAUGE	SAMPLING / HOLD
5.1	MRS Mirror Mount Seat Flatness	**≤ 0.015 mm across titanium flange**	Optical Flat & Monochromatic	Muzzle Tip Dovetail
5.2	Muzzle Telescope Optical Zero	**≤ 0.02 mrad rotation error (180°)**	Dioptometer & Collimator	4-Quadrant Roll Check
5.3	MRS Transceiver Return Signal	**≥ 85% photodiode intensity peak**	FCS Diagnostic Telemetry	Azimuth & Elevation Channels
5.4	Induced Bend Compensation Accuracy	**≤ 0.03 mrad residual error**	Calibrated Wedge Prism Test	± 1.5 mrad Simulated Bend
5.5	MRS Mirror Fastener Lock Torque	**32.0 Nm ± 1.5 Nm + wire lock**	Calibrated Torque Wrench	4 Titanium Retaining Bolts

HULL / TURRET SERIAL NO.	SHIFT / BAY CODE	OPERATOR STAMP	QA INSPECTOR STAMP	DISPOSITION
HVF-2026-135	SHIFT-A / GC-518	OP-772 (Verified)	DGQA-10 (Cleared)	**ACCEPTED — PASS**

STAGE-GATE WARNING (SECTION 5.0): Do not advance the armoured assembly to the next station until all five parameters in Table 5.0 are counter-stamped in the hull e-Passport ledger.

SECTION 6.0 — FIRE CONTROL SYSTEM (FCS) OPTICAL/THERMAL SIGHT & LRF HARMONIZATION

Operational Execution Protocol: On the 100-meter indoor collimated boresight range, the mechanical bore axis is harmonized with the Gunner's Primary Sight (GPS) day channel, cooled thermal imager (TI), Commander's Panoramic Sight, and eye-safe Laser Rangefinder (LRF). During shop-floor execution of *Fire Control System (FCS) Optical/Thermal Sight & LRF Harmonization*, the designated chargehand shall verify that calibration certificate **CAL-HVF-1611** and material traceability record **LOT-DL-0156** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, veritatis doloribus quaerat ex qui natus nemo totam id voluptas sed eveniet amet similique facere, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

Quality Assurance & Non-Conformance Control: Any deviation exceeding the specified tolerance band in Table Fire must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, odio numquam laborum maxime deleniti reprehenderit nulla dolor totam sit, while automated telemetry from station **CAL-HVF-1907** is archived against hull serial dossier **LOT-NE-4394** prior to releasing the assembly to the subsequent production workstation.

6.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **6.0** on platform *125mm Smoothbore 2A46M-5 / 120mm Ordnance System*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Architecto ea molestias nihil neque consectetur velit eos. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

TABLE 6.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

STEP	TECHNICAL PARAMETER / CHECKPOINT	ACCEPTANCE TOLERANCE	METROLOGY / GAUGE	SAMPLING / HOLD
6.1	Gunner Day Channel Coincidence	**≤ 0.05 mrad to mechanical bore**	100m Collimated Target Board	Azimuth & Elevation Zero
6.2	Thermal Imager (TI) Hot-Spot Zero	**≤ 0.06 mrad to day reticle**	Thermal Blackbody Crosshair	WFOV & NFOV Channels
6.3	Laser Rangefinder (LRF) Beam Axis	**≥ 90% energy within 0.25 mrad box**	IR Laser Beam Profiler Camera	Boresight Target Matrix
6.4	Commander Sight Slave Tracking	**≤ 0.10 mrad hunter-killer handoff**	Dual Sight Resolver Readout	8 Azimuth / 3 Elev Angles
6.5	Ballistic Computer Zero Store	**Non-volatile EEPROM checksum OK**	FCS Built-In Test (BIT) Log	Final DGQA Seal & Lock

HULL / TURRET SERIAL NO.	SHIFT / BAY CODE	OPERATOR STAMP	QA INSPECTOR STAMP	DISPOSITION
HVF-2026-049	SHIFT-A / GC-518	OP-180 (Verified)	DGQA-53 (Cleared)	**ACCEPTED — PASS**

STAGE-GATE WARNING (SECTION 6.0): Do not advance the armoured assembly to the next station until all five parameters in Table 6.0 are counter-stamped in the hull e-Passport ledger.

APPENDIX A — NORMATIVE DEFENCE STANDARDS, CALIBRATION & REVISION LEDGER

This Appendix establishes the normative statutory, metallurgical, and metrological references governing **Main Gun Barrel Fitment & Muzzle Reference Calibration** (HVF/AVNL/SOP-MBT/GC-518) at Heavy Vehicles Factory (HVF), Avadi. All instrumentation utilized in Sections 1.0 through 6.0 must maintain unbroken calibration traceability to National Physical Laboratory (NPL) / NABL standards. Suscipit adipisci necessitatibus saepe quos et animi doloribus eos minima culpa totam est.

STD / CODE	STANDARD TITLE & DEFENCE MANUFACTURING APPLICABILITY	CUSTODIAN
DGAQA / DGQA-ARM-09	Quality Assurance Norms for Combat Vehicle Hull, Turret & Running Gear	DGQA (MoD)
MIL-DTL-12560 / DMR	Armor Plate, Steel, Wrought, Homogeneous & High-Hardness Ballistic Spec	DMRL / HVF Met Lab
ISO 10012 / NABL-141	Measurement Management Systems — Metrological Confirmation & Calibration	HVF Metrology Cell
AVNL-HSE-SOP-014	Industrial Safety, Crane Rigging, Fume Extraction & Confined Space Entry	HVF Safety Dept

APPENDIX B — MULTI-TIER OFFICIAL SIGN-OFF & AUTHORIZATION MATRIX

This Standard Operating Procedure is officially released for defence production upon execution of the four-tier authorization matrix below by designated HVF, AVNL, and DGQA authorities at Avadi, Chennai.

TIER / ROLE	DESIGNATION & DIRECTORATE	OFFICER NAME & ID	DIGITAL STAMP / SIGNATURE	DATE
TIER 1: PREPARED BY	Deputy General Manager (Production Engg) HVF Avadi	Er. R. Venkataraman IOFS-4418	**[DIGITALLY SIGNED] DGM/PE/HVF-AVADI**	30-Oct-2026
TIER 2: REVIEWED BY	Joint General Manager (Quality & Metallurgy) HVF Avadi	Dr. S. Muthukumar IOFS-3802	**[DIGITALLY SIGNED] JGM/QA-MET/HVF**	30-Oct-2026
TIER 3: CONCURRENCE	Controller, Directorate General of Quality Assurance (DGQA Armoured Vehicles)	Brig. A. K. Rathore CQA(HV) Avadi	**[SEAL & COUNTERSIGNED] CQA(HV)/MOD/GOI**	30-Oct-2026
TIER 4: APPROVED BY	Chief General Manager / Executive Director Heavy Vehicles Factory (AVNL)	Shri K. Parthasarathy CGM / HVF Avadi	**[APPROVED FOR RELEASE] CGM-OFFICE/AVNL**	30-Oct-2026

CONTROLLED COPY WARNING: Master copy resides in the HVF Avadi Central Technical Archive (PLM Vault). Printed copies are uncontrolled after 30 days from issue date.

Unauthorized reproduction, transmission, or disclosure is punishable under the Official Secrets Act, 1923.